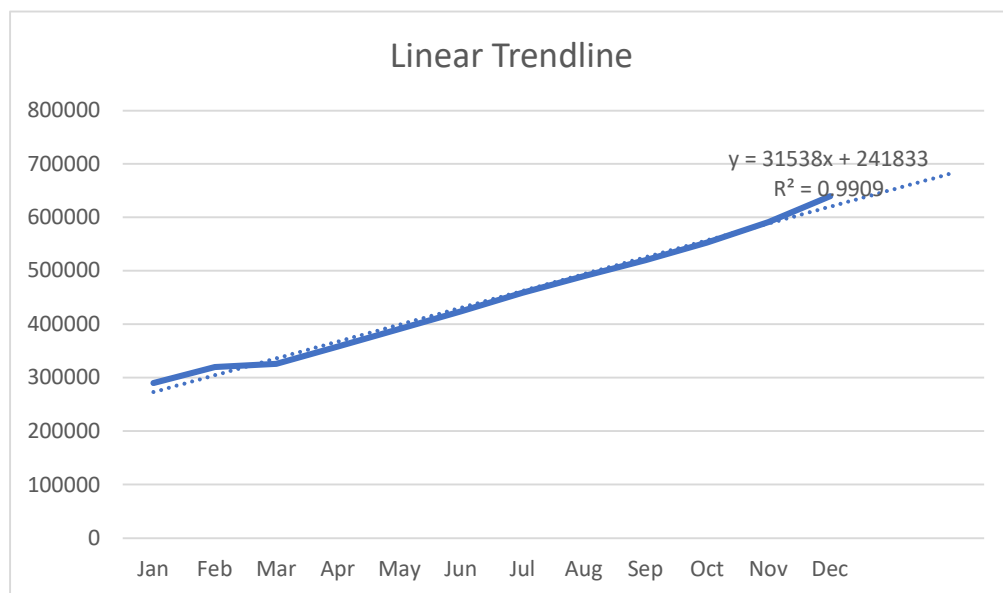


Row Labels	Sum of Sales (₹)
Jan	290000
Feb	320000
Mar	326000
Apr	358000
May	391000
Jun	424000
Jul	459000
Aug	490000
Sep	520000
Oct	553000
Nov	591000
Dec	640000
Grand Total	5362000



Problem 4 — Add a Linear Trendline + Forecast Business Requirement

Management wants to determine whether the company's monthly sales show a consistent upward trend and estimate

1. Is the overall sales trend increasing?

Yes, very clearly. Sales increased from: January → ₹290,000 to December → ₹640,000 That's an increase of: ₹640,000 – ₹290,000 = ₹350,000 Percentage increase: (640000–290000 / 290000) ×100 ≈ 120.7%	Why do we need to find this? <u>Because a Data Analyst shouldn't just report:</u> "December sales were ₹640,000." That's only a descriptive fact. <u>The more useful question is:</u> "What direction is the business moving?" The Line Chart immediately shows that sales are moving upward. <u>Business meaning</u> Management can conclude: "Sales have demonstrated strong and consistent growth throughout the year." <u>This can support decisions around:</u> Inventory planning , Sales targets , Staffing , Marketing budgets Revenue forecasting
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2. What is the Trendline Equation?

When you add a Linear Trendline in Excel and select Display Equation on chart, Excel will give approximately:	What does 31,538.46 mean? This is the most important part. <u>The slope is approximately:</u>
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$y = 31,538.46x + 241,833.33$ <u>Where:</u> x = time period/month number y = predicted sales $31,538.46$ = slope $241,833.33$ = intercept <u>So the equation is:</u> Sales = $31,538.46 \times \text{Month} + 241,833.33$	₹31,538 per month <u>So, according to the linear trend:</u> Sales are increasing by approximately ₹31.5K per month on average. <u>Why do we need the equation?</u> Because the chart shows the visual trend, but the equation gives us a mathematical model. <u>Instead of saying:</u> "The line seems to be going upward." <u>we can say:</u> "The fitted linear trend indicates an average increase of approximately ₹31,538 in sales per month." That's much more useful for analysis and forecasting.
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3. What is the R^2 Value?	
For this dataset: $R^2 \approx 0.991$ or approximately: 99.1%	
<u>In Excel, you'll see something close to: $R^2 = 0.991$</u>	
Why do we need R^2 ? This is where many beginners make a mistake. <u>They see:</u> $R^2 = 0.991$ and <u>think:</u> "Great, that's just another number." <u>No.</u> R^2 tells us how well the linear trendline fits the observed data. <u>Think of it as:</u> How well does the straight-line model explain the variation in sales? <u>Here:</u> $R^2 \approx 0.991$ <u>That's extremely high.</u> It means approximately 99.1% of the variation in the sales data is explained by the linear relationship with time in this dataset. <u>Interpretation</u> R^2 = General interpretation Near 0 = Very weak linear fit Around 0.5 = Moderate fit Around 0.7–0.8 = Stronger fit Near 1.0 = Very strong linear fit	Why is this useful? Suppose you have two datasets: Company A : $R^2 = 0.25$ Company B : $R^2 = 0.99$ Both might have increasing sales. But Company B has a much more consistent linear pattern, so a linear trendline is a much better representation of its historical behavior. Therefore, R^2 helps us judge: "Is this trendline actually a good representation of the historical data?"

Our: $R^2 = 0.991$. It means the sales data follows a very strong linear upward pattern.

4. How strong is the relationship between time and sales?

From our data:

Very strong positive relationship.

Why?

Because:

- Sales generally increase as the months progress.
- The trendline slopes upward.
- $R^2 \approx 0.991$.

You can explain this in an interview like this:

"The monthly sales data shows a very strong positive linear relationship with time. The R^2 value of approximately 0.991 indicates that the linear trend explains about 99.1% of the variation in historical sales."

That's a much better Data Analyst answer than simply saying: "Sales are increasing."

5. What sales does the trendline forecast for the next two months?

This is where the equation becomes useful.

We have: $y = 31,538.46x + 241,833.33$

Our data has 12 months.

So:

Jan = $x = 1$

Feb = $x = 2$

...

Dec = $x = 12$

Therefore:

Next month $\rightarrow x = 13$

$y = 31,538.46(13) + 241,833.33 \approx ₹651,833$

So:

January next year $\approx ₹651,833$

Following month $\rightarrow x = 14$

$y = 31,538.46(14) + 241,833.33 \approx ₹683,372$

So: February next year $\approx ₹683,372$